

# A Systematic Literature Review of Explainable Artificial Intelligence (XAI) for Interpreting Student Performance Prediction in Computer Science and STEM Education


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## Abstract

Educational Data Mining (EDM) supports early detection of learning difficulties by predicting student performance. However, machine learning models often operate as black boxes. Explainable Artificial Intelligence (XAI) helps to explain why black-box models produce specific predictions. This paper systematically reviews the past five years of research on XAI applications for interpreting student performance prediction in Computer Science and STEM education. We found that behavioral and academic performance data were the most commonly used features, with the main prediction goals focused on course failure risk or grades. This study also examined the application areas of XAI, revealing that the most common uses were global feature importance analysis, individual prediction explanations, and supporting interventions and decision-making. Moreover, we found that SHapley Additive exPlanations (SHAP) were the most frequently utilized XAI technique, predominantly applied at the global level, with limited use at the individual level. Furthermore, a research gap was identified in utilizing XAI to support course improvements, customize visualizations, and generate personalized recommendations. Addressing this gap could enable educators to provide personalized, data-driven guidance to better support individual students.



## CCS Concepts

• **Computing methodologies** → ***Artificial intelligence***
• **Applied computing** → ***Education***



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## Keywords

Explainable Artificial Intelligence, XAI, Educational Data Mining, EDM, Performance Prediction, Computer Science Education, STEM Education, Systematic Literature Review

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## 1 Introduction

Computer Science and STEM-related skills have become increasingly essential in the 21st century. However, these disciplines often present difficulties for learners, requiring a combination of mathematical operations, abstract concepts, programming syntax, and even language comprehension, which may challenge beginners [1] [2].

Educational Data Mining (EDM) is a branch of data mining that analyzes educational data using machine learning, deep learning, or statistical methods. One application of EDM is predicting student performance [3]. This supports early detection of learning problems [4], allowing pedagogical interventions that may guide students toward success [5]. However, traditional machine learning and deep learning models are typically black boxes, making it challenging to understand why they produce specific predictions.

Explainable Artificial Intelligence (XAI) techniques interpret student performance prediction models, helping educators understand factors contributing to student success or failure [6] [7]. XAI methods make AI model decisions more interpretable, enhancing model transparency and potentially improving decision-making processes [8] [9].

This systematic literature review explored the state-of-the-art XAI techniques in interpreting student performance predictions in Computer Science and STEM education. The reason for emphasizing these educational fields is their significant influence

on future technological development. The following research questions guided this study:

(1) What datasets, feature types, prediction objectives, and algorithms were used in XAI studies for interpreting student performance?

(2) What XAI techniques, interpretability levels, visualization methods, and practical implementations were employed in these studies?

The review addressed these questions to identify possible research gaps in the literature and suggest directions for improving the interpretability and utility of performance prediction models in Engineering education.

## 2 Related Work

In the reviewed literature, researchers usually use interpretability and explainability in XAI interchangeably, referring to the degree to which a human can understand why the AI model made a specific prediction [8] [10]. However, few studies considered interpretability and explainability as distinct concepts [11], in which interpretability refers to the inherent transparency of a model. In contrast, explainability emphasizes post-hoc techniques to elucidate model decisions.

Early reviews in the field of XAI aimed to define core interpretability concepts and classify XAI techniques. Adadi and Berrada [9] conducted a significant literature review in 2018, covering 381 papers published from 2004 to 2018. They classified XAI methods based on their applicability across machine learning algorithms, interpretability level (local or global), and interpretability timing (intrinsic or post-hoc). Intrinsic interpretability refers to implicitly explainable models due to their relatively simple structure, such as linear regression. Post-hoc interpretability involved applying additional techniques to explain the decisions of complex, black-box models.

Guidotti et al. [12] conducted a review in 2019 to categorize XAI methods into those for explaining black-box models, inspecting models, explaining their outcomes, and building inherently transparent models. They noted the absence of formal evaluation metrics for interpretability and the challenges of interpreting models relying on unknown or latent features.

Arrieta et al. [13] provided a review in 2020, focusing on distinguishing between transparent and post-hoc methods and identifying specific subcategories for deep learning interpretability techniques.

In 2023, Chaushi et al. [14] reviewed XAI in education, emphasizing the need for transparency to build trust. They identified challenges with complex AI models and interpretability requirements and suggested areas for future research relevant to XAI in education.

However, while existing reviews provided valuable insights into XAI, they primarily focused on the technological side rather than applying XAI in specific educational fields.

Our systematic literature review addressed this gap by analyzing recent research from the last five years (2020-2024) on applying XAI techniques in EDM for Computer Science and STEM education. We aimed to provide insights, identify research gaps, and propose future directions.

## 3 Methodology

This systematic literature review followed Kitchenham's framework [15] for conducting systematic reviews in Engineering or Computer Science education.

### 3.1 Search Strategy

We conducted a comprehensive literature search using the most popular Computer Science and Engineering research databases, ACM Digital Library, IEEE Xplore, and Web of Science.

The search string was constructed around four key concepts: educational data mining, XAI techniques, performance prediction, and programming or STEM education. After several refinement iterations, we formulated the search string:

("artificial intelligence" OR "AI" OR "machine learning" OR "deep learning" OR "educational data mining" OR "EDM") AND ("SHAP*" OR "LIME" OR "Local Interpretable Model-agnostic Explanations" OR "PDP" OR "Partial Dependence Plot" OR "Anchors" OR "ICE" OR "Individual Conditional Expectation" OR "ALE" OR "Accumulated Local Effect") AND ("explain*" OR "interpret*" OR "transparen*") AND ("performance") AND ("predict*") AND ("program*" OR "CS1" OR "computer" OR "STEM") AND ("education" OR "teach*" OR "school" OR "student" OR "class*" OR "academic" OR "course" OR "curriculum")

### 3.2 Inclusion and Exclusion Criteria

We established inclusion and exclusion criteria to ensure the selected papers aligned with our research questions.

The following criteria were defined for paper inclusion: i) studies that predict student performance in Computer Science or STEM education using machine learning or deep learning approaches; ii) studies that explicitly apply XAI techniques to interpret prediction models; iii) studies that have included explicit descriptions of XAI implementation methods; iv) studies published in peer-reviewed journals or conferences.

The exclusion criteria were: i) studies focused solely on analyzing student behavior without performance prediction; ii) research unrelated to Computer Science or STEM education; iii) papers not in English; iv) Position papers, reviews, or theoretical works without empirical results.

### 3.3 Search and Selection Process

As shown in Figure 1, we used the four stages of the search and selection process to identify the most relevant literature for subsequent analysis.

Stage 1: In July 2024, we applied the pre-defined search string to the abstracts and full texts across the databases, yielding 1,866 papers.

Stage 2: Removing 24 duplicates based on title, authors, and DOI left 1,842 papers.

Stage 3: We manually reviewed titles, abstracts, and keywords to check for inclusion and exclusion criteria, then conducted a text

search for key phrases. These relevant paragraphs were read to determine the role of XAI in the papers. This step allowed us to exclude studies that only mentioned XAI as related works but did not specifically cover XAI for interpreting student performance prediction. After this, 1,805 papers were excluded, leaving 37 papers.

Stage 4: Full-text screening of the remaining papers led to extracting detailed metadata (e.g., title, objectives, findings, educational context) and further exclusions based on relevance. This final step left 13 papers [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28] for analysis.

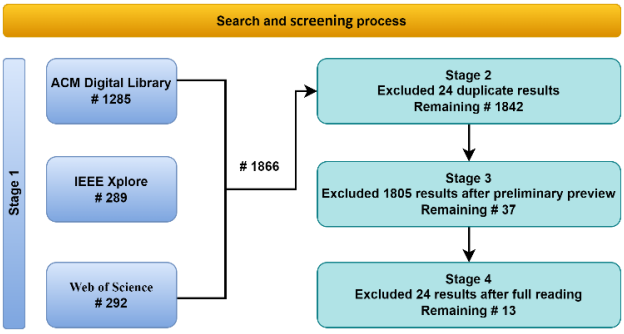



**Figure 1: Search and selection process**

# 4 Results for RQ1: The Datasets, Features, Prediction Objective, and Algorithms

## 4.1 Datasets and Data Sources

For datasets, as shown in Table 1, 69.2% of studies used self-collected datasets, while 30.8% employed the STEM-related Open University Learning Analytics Dataset (OULAD) [29].

**Table 1: Distribution of Datasets and Data Sources**

| Source | Availability | N | Percent | Papers |
|---|---|---|---|---|
| LMS | Self-collected | 4 | 30.8% | [16], [18], [27], [28] |
| | Public | 4 | 30.8% | [20], [22], [24], [26] |
| AJS | Self-collected | 2 | 15.4% | [21], [23] |
| SIS | Self-collected | 2 | 15.4% | [17], [19] |
| Others | Self-collected | 1 | 7.7% | [25] |

Regarding data sources, Learning Management Systems (LMS) were the most common data source, used in 61.5% of studies, followed by Auto Judge Systems (AJS) and Student Information Systems (SIS) at 15.4% each, with 7.7% relying on surveys and instructor observations [25].

Sample sizes ranged from under 100 to over 30,000 students, with larger datasets like OULAD enabling high prediction accuracy (e.g., 91.9% by Adnan et al. [20]). Smaller datasets also provided valuable insights. For example, Rico-Juan et al. [23] achieved 0.70 AUC with data from 90 students.

## 4.2 Feature Types and Input Variables

Predictive models are built on different features, which serve as input data. As shown in Table 2, each feature type includes specific input variables.

We found that feature diversity varies across reviewed papers, making it challenging for educators to gain clear insights from black-box models. XAI techniques are essential for clarifying feature contributions and supporting meaningful interpretation.

Behavioral data used in 29.7% of studies includes variables like clickstream data and participation data. For example, Swamy et al. [16] predicted course failure risk in a Scala programming course by tracking video-watching and quiz activities. Waheed et al. [26] predicted pass or fail outcomes in a STEM course based on Moodle interactions.

Academic performance data, used in 27.0% of studies, like assessment performance and course completion status, provides concrete measures of success. For instance, Adnan et al. [20] tracked assessment scores to classify student outcomes as Distinction, Pass, Fail, or Withdrawn.

Demographic data (16.2%) provides a broader context with variables such as age and gender. Programming data (13.5%) includes variables like code and syntax errors. Contextual data (13.5%) encompasses variables like course difficulty, credits, and activity patterns.

Among these, behavioral and academic performance features are not only the most frequently used but also the most crucial for interpretability in XAI applications, as they directly reflect students' engagement and achievement.

**Table 2: Feature Types and Input Variables**

| Type | Input Variables | Percent | Papers |
|---|---|---|---|
| Behavioral | Course clickstream data, textual comments, participation, attendance | 29.7% | [16], [18], [19], [20], [21], [22], [23], [24], [25], [26], [28] |
| Academic performance | GPA, course completion status, assessment scores | 27.0% | [16], [17], [19], [20], [21], [22], [24], [25], [26], [28] |
| Demographic | Gender, age, highest education level | 16.2% | [17], [19], [20], [22], [23], [24] |
| Programming data | Code, keystroke latency, programming constructs, syntax errors | 13.5% | [21], [23], [25], [27], [28] |
| Contextual | Course difficulty, credits, activity patterns | 13.5% | [16], [17], [19], [23], [25] |

## 4.3 Prediction Objectives and Algorithms

Understanding the prediction objectives is crucial, as they provide context that XAI techniques aim to explain. As shown in Table 3, a clear relationship exists between prediction objectives and algorithms.

The most prevalent objective is the Risk of Course Failure, as 61.5% of studies predict whether students will pass, fail, or drop out. Classification algorithms, such as Random Forest [30], XGBoost [31], and LSTM [32] are most widely used. For instance, Lu et al. [24] achieved 90% accuracy in predicting dropout using XGBoost in a STEM course.

**Table 3: The Reviewed Studies of XAI Techniques Applied in Student Performance Prediction**

| Authors | Prediction Objective | Target Variables | Best Algorithms | XAI Techniques | Levels | Visualization Methods |
|---|---|---|---|---|---|---|
| Swamy et al. [16] | Risk of Course Failure | Pass or fail in Scala programming course | BiLSTM with 90% accuracy | SHAP, LIME | Global | 1. XAI Method Similarity Heatmap<br>2. Feature Importance Heatmap |
| Baranyi et al. [17] | Risk of Course Failure | Graduate or drop out of STEM course | FCNN with 72.4% accuracy | SHAP, Permutation Importance [35] | Global | 1. SHAP Summary Plot<br>2. Permutation Importance Table |
| Alrajhi et al. [18] | Student Engagement | Comment urgency in Big Data course | BERT with 92% accuracy | Integrated Gradients | Individual | VisualizationDataRecord (Captum library [36]) |
| Mendez et al. [19] | Scores or Grade | Grade in CS course | GBT (accuracy not specified) | SHAP | Individual | Customized Pie charts (Show feature contributions via SHAP values) |
| Adnan et al. [20] | Risk of Course Failure | Outcome: Distinction, Pass, Fail, Withdrawn in STEM course | Random Forest with 91.9% accuracy | SHAP | Global, Individual | 1. Global (Summary Plot, Interaction Dependence Plot)<br>2. Individual (Local explainability) |
| Pereira et al. [21] | Risk of Course Failure | Pass or fail in Python course | XGBoost with 81.3% accuracy | SHAP | Global, Individual | 1. Global (Bar Plot, Summary Plot)<br>2. Individual (Decision Plot, Force Plot) |
| Rahul and Katarya [22] | Risk of Course Failure | Pass or fail in STEM course | EDL_KDF with 99.6% accuracy | SHAP | Global | Bar Plot, Summary Plot |
| Rico-Juan et al. [23] | Risk of Course Failure | Assignment pass or fail in Programming Challenges course | Random Forest with AUC 0.70 | SHAP | Global | Summary Plot, Dependence Plot |
| Lu et al. [24] | Risk of Course Failure | Drop out or not in STEM course | XGBoost with 90% accuracy, AUC-ROC 0.89 | SHAP | Global, Individual | 1. Global (Summary Plot, Bar Plot, Dependence Plot, Interaction Plot, SHAP Heatmap)<br>2. Individual (Waterfall Plot) |
| Giannakas et al. [25] | Scores or Grade | Team performance: A or F in Software Engineering course | DNN with 82.39% accuracy | SHAP | Global | Bar Plot, Summary Plot |
| Waheed et al. [26] | Risk of Course Failure | Pass or fail in STEM course | LSTM with 84.57% accuracy, AUC 0.82 | SHAP | Global | Summary Plot, Dependence Plot |
| Effenberger and Pelánek [27] | Code Understanding | Clusters of student code solutions in Python course | Custom clustering algorithm | Custom interpretable clustering algorithm | Global | Clustered feature matrix, Cluster summary diagram |
| Afzaal et al. [28] | Scores or Grade | Assignments performance in programming course | ANN with 90% accuracy | Counterfactual Explanations | Individual | Feedback Dashboard (DICE library [37]) |

Boosting algorithms or neural networks, such as Gradient-Boosting Trees (GBT) [33], Artificial Neural Networks (ANN) [34], and Deep Neural Networks (DNN) [38], are used to predict Scores or Grades in 23.1% of the studies. For example, Mendez et al. [19] used GBT for regression to predict continuous grade outcomes in a Computer Science course.

In the Code Understanding category, Effenberger and Pelánek [27] used clustering to group student programming solutions based on code features in the Python course.

For Student Engagement, Alrajhi et al. [18] applied text mining with Bidirectional Encoder Representations from Transformers [39] (BERT-based neural network) to classify the urgency of student comments in the Big Data course, achieving a high accuracy of 92%.

# 5 Results for RQ2: XAI Techniques

## 5.1 Type of XAI Techniques Used

As shown in Table 3, various XAI techniques have been applied to improve the interpretability of predictive models.

*5.1.1 SHAP.* As the most used technique, Shapley Additive Explanations (SHAP) [40] appears in 11 studies. For example, Pereira et al. [21] used SHAP with an XGBoost model to identify key student behaviors, helping educators understand which factors promote success and enabling targeted support for at-risk students in a Python course. SHAP's popularity can be attributed to its strong theoretical foundation in Game Theory [41], ability to provide local and global explanations, flexibility to work with most machine learning models, and accessibility through open-source libraries.

*5.1.2 LIME.* Only one study applied Local Interpretable Model-agnostic Explanations (LIME) [42], a widely used technique that explains individual predictions by creating simplified models around each instance. LIME constructs local surrogate models, such as linear regressions or decision trees, to approximate the complex model's behavior for specific predictions. Specifically, Swamy et al. [16] used LIME to identify critical predictors like quiz submissions and video engagement influencing success.

*5.1.3 Permutation Importance.* Applied in one study, Permutation Importance [35] assesses feature importance by shuffling each feature's values and measuring the effect on model accuracy. A significant accuracy drop indicates a more important feature. Baranyi et al. [17] used it to validate predictors for graduation, confirming factors like university entrance scores as critical indicators in a STEM course.

*5.1.4 Integrated Gradients.* Alrajhi et al. [18] adopted Integrated Gradients [43], which compute gradients along paths to features, to explain a BERT model for urgent student comments. This technique highlighted keywords, aiding educators in identifying

areas needing prompt responses and improving model transparency.

*5.1.5 Counterfactual Explanations.* In Afzaal et al. [28], Counterfactual Explanations [44] were employed to create what-if scenarios by adjusting feature values. This approach explores how minimal changes in input features could impact model predictions. This study aims to leverage personalized recommendations to empower students in a programming course, enabling them to take targeted actions to enhance their performance.

*5.1.6 Custom Interpretable Clustering Algorithm.* Effenberger and Pelánek [27] employed a custom interpretable clustering algorithm in a Python course to categorize student code submissions based on standard programming patterns, such as loops and conditional statements. This approach enabled educators to quickly identify shared coding behaviors, facilitating more targeted feedback and enhancing the interpretability of performance data in programming tasks.

While various XAI techniques were applied, methods such as Accumulated Local Effects (ALE) [45], Anchors [46], and Deep Learning Important Features (DeepLIFT) [47] were notably absent in the reviewed studies. This gap presents an opportunity for future research to incorporate these techniques into student performance prediction models.

## 5.2 Practical Applications of XAI

We found no existing literature reviews focused on practical XAI applications in Computer Science and STEM education. To address this gap, we summarized relevant applications that may support educational outcomes in Table 4.

**Table 4: XAI Applications in CS and STEM Education**

| Category | Percent | Papers |
|---|---|---|
| Global feature importance analysis | 33.3% | [16], [17], [20], [21], [22], [23], [24], [25], [26], [27] |
| Individual prediction explanation | 20% | [18], [19], [20], [21], [24], [28] |
| Supporting interventions and decision-making | 26.7% | [18], [19], [20], [21], [24], [25], [26], [27] |
| Feature interaction exploration | 13.3% | [20], [23], [24], [27] |
| Actionable course improvement | 3.3% | [16] |
| Generate recommendation | 3.3% | [28] |

*5.2.1 Global Feature Importance Analysis.* A primary application, accounting for 33.3% of reviewed studies, is global feature importance analysis, which aims to identify features most relevant to overall student performance. For instance, Pereira et al. [21] used SHAP summary plots to analyze critical predictors of student success globally, identifying the first exam grade and time spent accessing the system as crucial factors for academic achievement in the CS1 course.

*5.2.2 Individual Prediction Explanation.* Covering 20% of studies, individual prediction explanations provide targeted insights for individual students by highlighting influential behaviors. For example, Lu et al. [24] used the SHAP Waterfall Plot to show the contributions of features such as resource and assignment clicks, clarifying how these factors influence a learner's probability of continuing or dropping out of a STEM course.

*5.2.3 Supporting Interventions and Decision-Making.* In 26.7% of studies, XAI applications assist educators in making timely interventions to support student progress. For example, Alrajhi et al. [18] used Captum visualization [36] to help teachers identify comments requiring urgent attention, allowing for targeted support where it is most needed.

*5.2.4 Feature Interaction Exploration.* Comprising 13.3% of studies, feature interaction exploration examines how various factors jointly influence student outcomes. For instance, Waheed et al. [26] used SHAP dependence plots to explore the relationship between quiz engagement and course assignments. They revealed that consistent participation in both activities correlates with a lower risk of failure, underlining their combined positive impact on student performance.

*5.2.5 Actionable Course Improvement.* Only one of the reviewed studies used XAI to look for course improvements. Swamy et al. [16] conducted qualitative interviews, finding that over 85% of educators valued XAI insights for refining course elements in a programming course, such as adjusting quiz challenges.

*5.2.6 Generating Recommendations.* Also one of the reviewed studies generated recommendations focused on personalized feedback. Afzaal et al. [28] designed a dashboard offering tailored advice. For example, if quiz scores suggest an improvement, the dashboard generates recommendations for students on relevant study materials, helping students focus on practical strategies for improving their programming skills.

In summary, there is a notable gap in using XAI to generate personalized recommendations. Addressing this gap would allow educators to offer personalized, data-driven guidance to better support individual students. Moreover, XAI applications for course improvements are still limited, suggesting further opportunities to enhance teaching and optimize learning outcomes.

## 5.3 Levels of Interpretability

When examining XAI techniques for student performance prediction, interpretability methods are categorized into global and individual (local) levels, each offering distinct insights. Seven reviewed studies focus on global-level explanations, three on individual-level explanations, and three incorporate both.

*5.3.1 Global-Level.* Global interpretability provides insights into feature importance across all predictions, making it easier to observe dataset-wide patterns. In the reviewed studies, SHAP is commonly employed to identify feature importance and dependencies on a global level. While it provides comprehensive insights, SHAP requires high computational resources, especially with high-dimensional data, which can increase processing costs.

*5.3.2 Individual-Level.* Individual interpretability focuses on feature impacts for single predictions, supporting personalized insights for individual students. Techniques like Integrated Gradients [18] and Counterfactual Explanations [28] provide feature-specific contributions and minimal outcome-altering changes, enabling tailored interventions.

On the other hand, LIME appeared in one of the reviewed studies [16]. LIME offers more intuitive individual explanation visualizations than SHAP by generating a local surrogate model to approximate predictions. It can help users better understand how each feature contributes to the predicted outcome for a specific instance.

Most reviewed studies emphasize global explanations, though individual-level analysis remains crucial for personalized educational interventions. Moreover, Swamy et al. [16] highlighted that different XAI methods, such as SHAP and LIME, can yield different explained results on the same dataset. These challenges may explain why researchers used a single XAI method to maintain explanation consistency.

## 5.4 Model Performance and Interpretability

Balancing predictive accuracy and interpretability is essential when developing performance prediction models.

**Table 5: Machine Learning vs Deep Learning**

| Type | Percent | Algorithm | Model Accuracy | Best Algorithm |
|---|---|---|---|---|
| Traditional Machine Learning | 53.8% | GBT [19], ANN[28], Random Forest[20], [23], XGBoost[21], [24], Custom clustering [27] | 88.3% ± 4.8 | Random Forest with 91.9% accuracy [20] |
| Deep Learning | 46.2% | BiLSTM [16], FCNN [17], BERT [18], EDL_KDF [22], DNN [25], LSTM [26] | 86.8% ± 9.3 | EDL_KDF with 99.6% accuracy [22] |

*5.4.1 Traditional Machine Learning*. As shown in Table 5, the GBT, Random Forest, XGBoost, ANN, and custom clustering algorithms make up 53.8% of the reviewed approaches. These traditional machine learning models achieve an average accuracy of 88.3% ± 4.8, with Random Forest performing best at 91.9% [20]. These models employ XAI methods like SHAP and Counterfactual Explanations to enhance interpretability.

*5.4.2 Deep Learning*. Advanced architectures such as Bidirectional LSTM (BiLSTM), Fully Connected Neural Networks (FCNN), BERT, Explainable Deep Learning based Knowledge Distillation Framework (EDL_KDF), DNN, and LSTM represent 46.2% of the methods and achieve an average accuracy of 86.8% ± 9.3. This variability suggests that deep learning has the potential to achieve even higher performance. Although deep learning is traditionally viewed as less interpretable [48], this review found that various XAI techniques, such as SHAP, LIME, and Integrated Gradients, can be used to interpret deep learning models.

*5.4.3 Summary*. As deep learning model achieved the highest accuracy, 99.6%, using the EDL_KDF deep learning framework [22], which combined DAbDBi-LSTM and ResNet-101 networks. This framework outperformed traditional machine learning models. Therefore, if predictive accuracy is the primary goal and the interpretability challenges of deep learning models can be addressed, deep learning algorithms should be the first choice.

## 5.5 Visualization Methods

XAI visualization methods for student performance prediction can be divided into two main categories.

*5.5.1 XAI-Generated Plots*. This approach uses standard plots from libraries to visualize feature influence on model predictions, aiding educators and researchers in understanding complex models. Alrajhi et al. [18] used Captum's Integrated Gradients to highlight keywords affecting urgency classifications in comments. Meanwhile, Baranyi et al. [17] employed SHAP summary plots to illustrate how specific factors impact graduation probability. The reviewed papers demonstrated that visualizations created using XAI libraries in sci-kit-learn [23], TensorFlow [16], and PyTorch [28], enable faster development of interpretable solutions.

*5.5.2 Customized Visualizations*. This approach customizes XAI outputs for improved accessibility, facilitating interpretation by educators and students. Mendez et al. [19] created pie charts based on SHAP values to clarify feature contributions visually. However, few studies have explored this direction, revealing a gap in developing educator and learner-friendly visualization tools. Addressing this could enhance the accessibility and practical value of XAI techniques in educational settings.

# 6 Conclusion and Discussion

This systematic literature review explored the current use of XAI techniques for interpreting student performance predictions within Computer Science and STEM education.

For RQ1, most studies used self-collected datasets, primarily from LMS platforms. Behavioral and academic performance data were the most common features, with the main prediction goal focused on course failure risk. However, the variety of features made it harder for educators to interpret black-box models, underscoring the need for XAI techniques to support meaningful interpretation.

For RQ2, this study examined the application areas of XAI, revealing that the most common uses were global feature importance analysis, individual prediction explanations, and supporting interventions and decision-making. Moreover, we found that SHAP was the most widely used XAI technique, primarily applied globally, with limited use at the individual level. However, XAI methods such as ALE, Anchors, and DeepLIFT were notably absent, suggesting opportunities for further research. Furthermore, there was a notable gap in using XAI for course improvements, customized visualizations, and generating personalized recommendations. Deep learning models demonstrated better performance than traditional models, highlighting their potential when predictive accuracy is prioritized.

Future studies could enhance the practical impact of XAI in education by addressing these research gaps. This would enable educators to provide personalized guidance that better supports students in Computer Science and STEM education.

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